Adapting COMET Models for Quality Evaluation of Low-Resource Machine Translation

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Abstract

The COMET series of machine translation quality evaluation models is built upon the popular multilingual encoder XLM-RoBERTa (XLM-R) and thus claims support for the 100 languages in XLM-R's training data. However, expanding support for low-resource languages is essential. In this paper, we investigate how to best fine-tune COMET models to support additional languages. We find that fine-tuning the underlying encoder on monolingual data in addition to the COMET model is crucial to improve correlation with human Multidimensional Quality Metrics (MQM) annotations. It is possible to improve COMET's accuracy with low-resource languages like Haitian Creole, Kiribati, Marshallese, and Navajo without sacrificing accuracy on high-resource languages like Spanish and Portuguese. We find promising results with both COMET, the quality evaluation model, and COMET-QE, the reference-free quality estimation model. We hope that our findings contribute to equitable access to accurate machine translation for all underserved languages.

1 Introduction

Quality evaluation is the scoring of machine translation output with respect to a source text and a reference translation. Quality estimation (QE), on the other hand, scores translation output with respect to just the source text. Evaluation can either be performed with traditional precision-based metrics like BLEU (Papineni et al., 2002) and ChrF (Popović, 2015) or with neural metrics like COMET (Rei et al., 2020), while estimation is done primarily with neural metrics. Neural metrics have shown to correlate highly with human judgment, at least with high-resource language pairs. They rely on training data in the form of human quality assessments like Multidimensional Quality Metrics (MQM) annotations (Lommel et al., 2014). However, neural metrics often suffer for low-resource language pairs. QE is especially important among low-resource language pairs when machine translation output is particularly messy and inaccurate. In a high-stakes translation scenario, for example, language service providers could use QE models to discard questionable translations. QE models could be used as a training data filter to improve translation model quality (Chimoto and Bassett, 2022). Furthermore, a trustworthy metric is helpful for measuring improvements in model quality.

High-Resource	Low-Resource
Spanish (es)	Marshallese (mh)
Portuguese (pt)	Kiribati (gil)
Tagalog (tg)	Haitian Creole (ht)
Thai (th)	Navajo (nv)
Armenian (hy)	
Croatian (hr)	

Table 1: The languages whose translations from English are evaluated, sorted from most to least monolingual data available.

The problem with neural metrics and low-resource language pairs is that they often require a lot of training data to perform well. For this reason, traditional metrics are really powerful in the realm of quality evaluation. However, neural metrics are the predominant method for performing QE. Low-resource languages are less likely to have reference translations available, so QE is the more prevalent task in that domain. There is still potential for quality evaluation neural metrics to outperform traditional metrics on low-resource language pairs despite the need for training data, and there is plenty of room to grow in improving the quality of QE neural metrics.

One of the major drawbacks of the popular neural metric COMET is its reliance on the multilingual XLM-R encoder model which was only trained to support 100 languages (Rei et al., 2020; Conneau et al., 2020). Furthermore, the latest

COMET models themselves have only been fine-tuned on MQM data for a few high-resource language pairs (Rei et al., 2022). In this paper, we experiment with fine-tuning XLM-R on monolingual data from four unsupported low-resource languages—Haitian Creole, Kiribati, Marshallese, and Navajo—as well as some already supported major high-resource languages (see Table 1). We then incorporate the fine-tuned encoder model into the COMET and COMET-QE models. The COMET models are fine-tuned and evaluated on English to X translation pairs in a religious domain. We find that pre-training the encoder before fine-tuning the neural metric on parallel data greatly improves model correlation with human MQM annotations.

2 Related Works

Previous experiments with adapting COMET for unsupported languages have been conducted using a straight-forward fine-tuning technique. Falcão, et al. found that COMET performance for the translation pairs English-Maltese and Spanish-Basque increased by fine-tuning with COMET's recommended hyper-parameters but was susceptible to data distributions (Falcão et al., 2024). Their evaluation training data was collected through crowd-sourcing. They chose public BERT-based encoders that were previously fine-tuned with non-domain-specific Maltese and Basque data.

More recently, Wang, et al. conducted a similar experiment by assembling a dataset for 13 diverse under-resourced African languages (AfriMTE) and using it to create AfriCOMET, an African-centric quality evaluation model (Wang et al., 2024). AfriCOMET was based on AfroXLM-R, an African-centric multilingual encoder that leverages well-resourced language data. AfriMTE was created with machine translations from NLLB-200 and M2M-200. A reference-free QE version of AfriCOMET was also trained. AfriCOMET models demonstrated the possibility for transfer learning in quality evaluation models.

In this study, we diversify translation data quality by using a wider assortment of open machine translation models, since a balanced distribution of language data and quality appears to be crucial (Falcão et al., 2024). We select unique low-resource language pairs and create more quality annotation data for those pairs. We also focus in particular on the effects of encoder fine-tuning and try to maintain high-resource language pair performance.

Language	# Data
mh	266,681
gil	263,374
ht	197,914
es	78,520
en	77,969
pt	72,320
tg	32,719
th	25,158
nv	23,762
hy	18,180
hr	17,304

Lang. Pair	# Data
en-pt	9,198
en-tg	7,513
en-es	7,172
en-hy	6,692
en-ht	4,000
en-mh	3,000
en-gil	2,000
en-hr	2,000
en-nv	1,398
en-th	1,000

Table 2: The distribution of monolingual and parallel data used to train and evaluate the XLM-R and COMET models.

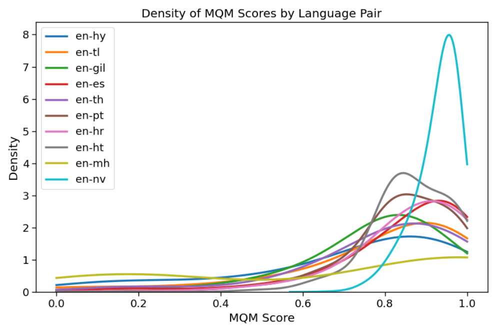

Figure 1: The distribution of MQM scores for the translation data of all language pairs.

3 Methodology

3.1 Datasets

Experiments are conducted using both FLoRes-200 (Team et al., 2022) and the proprietary datasets of The Church of Jesus Christ of Latter-day Saints due to their high multilinguality. The Church dataset contains both monolingual data and translation units. It is currently in the process of publication. We select six high-resource languages and four low-resource languages to experiment with. The high-resource languages in decreasing order of data are Spanish, Portuguese, Tagalog, Thai, Armenian, and Croatian. English is also included as it is the source language for all translation units. The low-resource languages in decreasing order of data are Marshallese, Kiribati, Haitian Creole, and Navajo. These languages were chosen due to their linguistic diversity as well as data and annotator availability.

Translation units from each language pair are selected to act as reference translations. The source from each of these translation units are

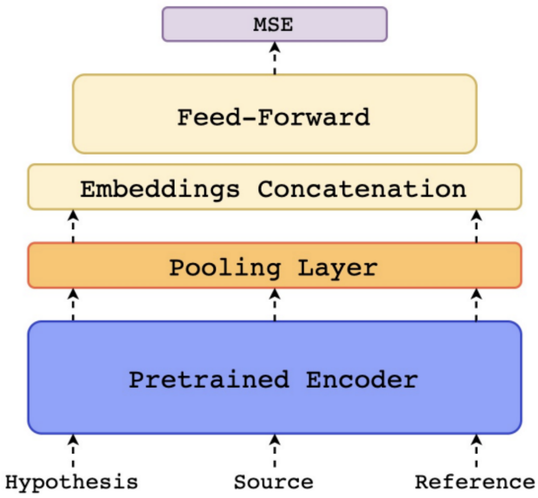

Figure 2: An architecture diagram of the COMET neural metric from the original paper (Rei et al., 2020).

then translated using a variety of machine translation systems to create a wide spread of translation quality: OpenNMT (Klein et al., 2017), NLLB-200 (Koishekenov et al., 2023), Microsoft Translate, Azure, ChatGPT, Google Translate, OPUS (Tiedemann et al., 2023; Tiedemann and Thottingal, 2020), IBM, MBART50 (Tang et al., 2020), T5 (Raffel et al., 2023), and Small100 (Mohammadshahi et al., 2022). Certain Spanish and Portuguese pairs are also translated by student translators working for BYU Speeches.

These machine translations along with some reference pairs are then given to BYU faculty to evaluate with the MQM framework. Evaluations are completed using the MQM Scorecard website developed in a collaboration between the German Research Center for Artificial Intelligence (DFKI) GMbH, the Brigham Young University Translation Research Group, and LTAC Global. The MQM scorecards are used to calculate error scores which are normalized between 0 and 1 for COMET fine-tuning. A large number of perfect scores are found, so they are down-sampled to create a more even distribution of scores (see Table 2 and Figure 1). There is of course some natural bias and variation among human annotators, which is one limitation of our data.

3.2 Model Architecture

Given a sentence embedding for the source, the hypothesis, and an optional reference as generated by a pre-trained cross-lingual encoder, COMET models extract combined features, concatenate them, and pass them through a feed-forward regressor. For this experiment, we use the `Unbabel/wmt22-comet-da` model (COMET) for quality evaluation and `Unbabel/wmt22-cometkiwi-da` model (COMET-QE) for quality estimation, since they are the latest and default models. These models are based on the XLM-RoBERTa encoder architecture.

In order to adapt these models to support our low-resource languages that are not supported by XLM-R, we first experiment with fine-tuning the base COMET and COMET-QE models on our MQM data. We then experiment with fine-tuning `XLM-RoBERTa-Large` on the Church's monolingual data, swapping out the base XLM-R with our fine-tuned XLM-R, and fine-tuning COMET and COMET-QE models on the same MQM data. There are six models evaluated in total: base COMET, fine-tuned COMET, fine-tuned COMET with XLM-R, base COMET-QE, fine-tuned COMET-QE, and fine-tuned COMET-QE with XLM-R.

3.3 Training Procedure

The COMET models are fine-tuned to minimize the mean squared error between the predicted scores and the MQM quality assessment scores. We choose the same training hyper-parameters as recommended by Unbabel. For COMET-QE, we use the Adam optimizer with 0.3 frozen epochs, a learning rate of 1.5e-05, a batch size of 16, layerwise decay of 0.95, Feed-forward hidden units of 2048 and 1024, Feed-Forward dropout of 0.1, and Feed-Forward activations with Tanh. For COMET, we use the same hyper-parameters except with an encoder learning rate of 1e-06 and Feed-forward hidden units of 3072 and 1024 to accommodate the reference data.

3.4 Model Evaluation

The MQM data is split into five folds for cross-validation with 80% for training data, 10% for test data, and 10% for validation data. Model output for the test data is evaluated by calculating Pearson correlation with MQM scores, a proxy for human evaluation. As a baseline for quality evaluation experiments, traditional precision-based metrics BLEU and ChrF and their correlation with MQM scores are also calculated. The Pearson correlation of each metric with MQM scores over the five folds of data are averaged to give a more accurate figure.

Metric	en-es	en-gil	en-hr	en-ht	en-hy	en-mh	en-nv	en-pt	en-th	en-tl	Avg
BLEU	0.208	0.108	0.255	0.123	0.300	0.439	0.230	0.187	0.031	0.226	0.211
ChrF	0.310	0.105	0.341	0.175	0.484	**0.806**	0.131	0.244	0.321	0.246	0.316
COMET	0.509	0.128	0.488	0.174	0.671	0.696	0.181	0.411	0.400	0.371	0.403
Ft COMET	0.501	0.153	**0.521**	0.172	**0.685**	0.739	0.266	**0.445**	**0.457**	**0.523**	0.446
Ft XLM-R+COMET	**0.517**	**0.192**	0.519	**0.216**	0.645	**0.806**	**0.332**	0.427	0.455	0.510	**0.462**
COMET-QE	0.496	0.061	0.476	0.134	0.649	0.282	0.303	0.404	0.378	0.320	0.350
Ft COMET-QE	0.497	0.131	0.501	0.022	**0.660**	0.209	0.216	0.446	0.509	0.546	0.374
Ft XLM-R+COMET-QE	**0.518**	**0.194**	**0.545**	**0.174**	0.636	**0.498**	**0.305**	**0.464**	**0.528**	**0.572**	**0.443**

Table 3: Average Pearson correlation of translation metrics with MQM scores by language pair across five folds of test data. Bolded values represent the best scores for each metric in the quality evaluation and quality estimation paradigms.

4 Results

Table 3 contains a summary of the results from all experiments. The most notable finding is that fine-tuned models exhibit the highest correlation with MQM scores. In particular, the models with a fine-tuned encoder perform the best among all low-resource language pairs. One exception is that the traditional metric ChrF correlates just as well as the fine-tuned XLM-R+COMET neural metric for English to Marshallese in the quality evaluation domain, perhaps due to Marshallese being a morphologically rich language that requires character-level precision. To see parity plots for each neural metric, refer to Appendix A.

4.1 Impact on High-Resource Pairs

The correlation of fine-tuned models with MQM scores of high-resource language pairs does not suffer from the inclusion of low-resource language data. For some high-resource language pairs, correlation drops a negligible amount after the fine-tuning of XLM-RoBERTa but remains high. In other words, it is possible to increase the correlation of low-resource language pairs without sacrificing the accuracy for high-resource language pairs when trained properly (i.e. with appropriately sampled training data).

4.2 Fine-Tuned Encoders

In the domain of quality estimation, some low-resource language pairs (Haitian Creole, Marshallese, and Navajo) see drops in performance even after fine-tuning on in-domain language data. These drops are counteracted when XLM-RoBERTa is also fine-tuned to support these languages. This trend highlights the importance of fine-tuning the encoder underlying neural metrics before fine-tuning the metric itself. A fine-tuned encoder provides more meaningful features to the feed-forward regressor as opposed to misleading features from an ill-informed encoder.

4.3 Effect of Training Data

Some increases in MQM score correlation are negligible. For quality estimation, English to Navajo experiences the smallest increase, but the language pair also has the least training data for both XLM-R and COMET of all the low-resource language pairs. Monolingual data for Navajo falls far behind, and its distribution of MQM scores is more concentrated toward perfect scores (see Figure 1). Regardless, English to Navajo also sees a sizable improvement in correlation for quality evaluation compared to both traditional metrics and the base neural metric COMET. Unlike English to Navajo, English to Marshallese has the most monolingual data and the most balanced distribution of MQM scores, and it sees the biggest improvement in correlation for QE. Perhaps increasing monolingual Navajo data to the levels of the other low-resource languages and constructing a translation corpus of more balanced MQM scores would yield a larger improvement in both quality evaluation and estimation.

5 Conclusion

In this paper we experiment with pre-training encoders to improve the efficacy of fine-tuning neural metrics for machine translation quality evaluation and estimation on previously unsupported language pairs. We demonstrate that encoder pre-training yields improvements in neural metric correlation with MQM scores for both high- and low-resource language pairs. In the domain of quality evaluation, neural metrics exceed the performance of traditional metrics like BLEU and ChrF. This technique may prove particularly useful if the amount of monolingual data available for a language pair in question is substantially larger than the amount of parallel data. Improved quality evaluation models for low-resource languages are crucial for the development of machine translation for those languages.

5.1 Future Work

As mentioned above, there are some limitations to our study. Namely, the distribution of English to Navajo translation data quality could be more balanced, and more monolingual Navajo data could be very beneficial. Another way to improve the quality of training data would be to have several annotators of the same language to reduce bias. It would be interesting to see if those data augmentations would make a substantial difference in model correlation with MQM scores.

Potential directions we would like to explore include expanding encoder vocabulary to include tokens specific to the unsupported languages. Additional language-specific tokens could make encoder output more meaningful for quality evaluation and estimation tasks.

We would also like to experiment with fine-tuning the latest XCOMET model, which performs both sentence-level evaluation and error span detection for highlighting problematic segments (Guerreiro et al., 2023). Our MQM data contains word-level quality evaluation annotations which are well-suited for fine-tuning XCOMET. The MQM data could also be used to create an automatic post-editing model, which would be novel for the low-resource languages we selected.

Of course, this experiment could be further replicated for other low-resource languages or domains. We selected only a small sample of languages in a very narrow domain. Our translation pairs are also English-centric.

References

Everlyn Chimoto and Bruce Bassett. 2022. COMET-QE and active learning for low-resource machine translation. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 4735–4740, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.

Júlia Falcão, Claudia Borg, Nora Aranberri, and Kurt Abela. 2024. COMET for low-resource machine translation evaluation: A case study of English-Maltese and Spanish-Basque. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 3553–3565, Torino, Italia. ELRA and ICCL.

Nuno M. Guerreiro, Ricardo Rei, Daan van Stigt, Luisa Coheur, Pierre Colombo, and André F. T. Martins. 2023. xcomet: Transparent machine translation evaluation through fine-grained error detection. *Preprint*, arXiv:2310.10482.

Guillaume Klein, Yoon Kim, Yuntian Deng, Jean Senellart, and Alexander M. Rush. 2017. Opennmt: Open-source toolkit for neural machine translation. *Preprint*, arXiv:1701.02810.

Yeskendir Koishekenov, Alexandre Berard, and Vassilina Nikoulina. 2023. Memory-efficient nllb-200: Language-specific expert pruning of a massively multilingual machine translation model. *Preprint*, arXiv:2212.09811.

Arle Lommel, Aljoscha Burchardt, and Hans Uszkoreit. 2014. Multidimensional quality metrics (mqm): A framework for declaring and describing translation quality metrics. *Tradumàtica: tecnologies de la traducció*, 0:455–463.

Alireza Mohammadshahi, Vassilina Nikoulina, Alexandre Berard, Caroline Brun, James Henderson, and Laurent Besacier. 2022. Small-100: Introducing shallow multilingual machine translation model for low-resource languages. *Preprint*, arXiv:2210.11621.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA. Association for Computational Linguistics.

Maja Popović. 2015. chrF: character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395, Lisbon, Portugal. Association for Computational Linguistics.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2023. Exploring the limits of transfer learning with a unified text-to-text transformer. *Preprint*, arXiv:1910.10683.

Ricardo Rei, José G. C. de Souza, Duarte Alves, Chrysoula Zerva, Ana C Farinha, Taisiya Glushkova, Alon Lavie, Luisa Coheur, and André F. T. Martins. 2022. COMET-22: Unbabel-IST 2022 submission for the metrics shared task. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 578–585, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Ricardo Rei, Craig Stewart, Ana C Farinha, and Alon Lavie. 2020. COMET: A neural framework for MT evaluation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2685–2702, Online. Association for Computational Linguistics.

Yuqing Tang, Chau Tran, Xian Li, Peng-Jen Chen, Naman Goyal, Vishrav Chaudhary, Jiatao Gu, and Angela Fan. 2020. Multilingual translation with extensible multilingual pretraining and finetuning. *Preprint*, arXiv:2008.00401.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, Anna Sun, Skyler Wang, Guillaume Wenzek, Al Youngblood, Bapi Akula, Loic Barrault, Gabriel Mejia Gonzalez, Prangthip Hansanti, John Hoffman, Semarley Jarrett, Kaushik Ram Sadagopan, Dirk Rowe, Shannon Spruit, Chau Tran, Pierre Andrews, Necip Fazil Ayan, Shruti Bhosale, Sergey Edunov, Angela Fan, Cynthia Gao, Vedanuj Goswami, Francisco Guzmán, Philipp Koehn, Alexandre Mourachko, Christophe Ropers, Safiyyah Saleem, Holger Schwenk, and Jeff Wang. 2022. No language left behind: Scaling human-centered machine translation. *Preprint*, arXiv:2207.04672.

Jörg Tiedemann, Mikko Aulamo, Daria Bakshandaeva, Michele Boggia, Stig-Arne Grönroos, Tommi Nieminen, Alessandro Raganato Yves Scherrer, Raul Vazquez, and Sami Virpioja. 2023. Democratizing neural machine translation with OPUS-MT. *Language Resources and Evaluation*, (58):713–755.

Jörg Tiedemann and Santhosh Thottingal. 2020. OPUS-MT — Building open translation services for the World. In *Proceedings of the 22nd Annual Conferenec of the European Association for Machine Translation (EAMT)*, Lisbon, Portugal.

Jiayi Wang, David Ifeoluwa Adelani, Sweta Agrawal, Marek Masiak, Ricardo Rei, Eleftheria Briakou, Marine Carpuat, Xuanli He, Sofia Bourhim, Andiswa Bukula, Muhidin Mohamed, Temitayo Olatoye, Tosin Adewumi, Hamam Mokayed, Christine Mwase, Wangui Kimotho, Foutse Yuehgoh, Anuoluwapo Aremu, Jessica Ojo, Shamsuddeen Hassan Muhammad, Salomey Osei, Abdul-Hakeem Omotayo, Chiamaka Chukwuneke, Perez Ogayo, Oumaima Hourrane, Salma El Anigri, Lolwethu Ndolela, Thabiso Mangwana, Shafie Abdi Mohamed, Ayinde Hassan, Oluwabusayo Olufunke Awoyomi, Lama Alkhaled, Sana Al-Azzawi, Naome A. Etori, Millicent Ochieng, Clemencia Siro, Samuel Njoroge, Eric Muchiri, Wangari Kimotho, Lyse Naomi Wamba Momo, Daud Abolade, Simbiat Ajao, Iyanuoluwa Shode, Ricky Macharm, Ruqayya Nasir Iro, Saheed S. Abdullahi, Stephen E. Moore, Bernard Opoku, Zainab Akinjobi, Abeeb Afolabi, Nnaemeka Obiefuna, Onyekachi Raphael Ogbu, Sam Brian, Verrah Akinyi Otiende, Chinedu Emmanuel Mbonu, Sakayo Toadoum Sari, Yao Lu, and Pontus Stenetorp. 2024. Afrimte and africomet: Enhancing comet to embrace under-resourced african languages. *Preprint*, arXiv:2311.09828.

A Parity Plots

Parity plots that compare human scores with automatic scores from all neural metrics are shown in Figure 3. Fine-tuning visibly narrows the range of automatic scores.

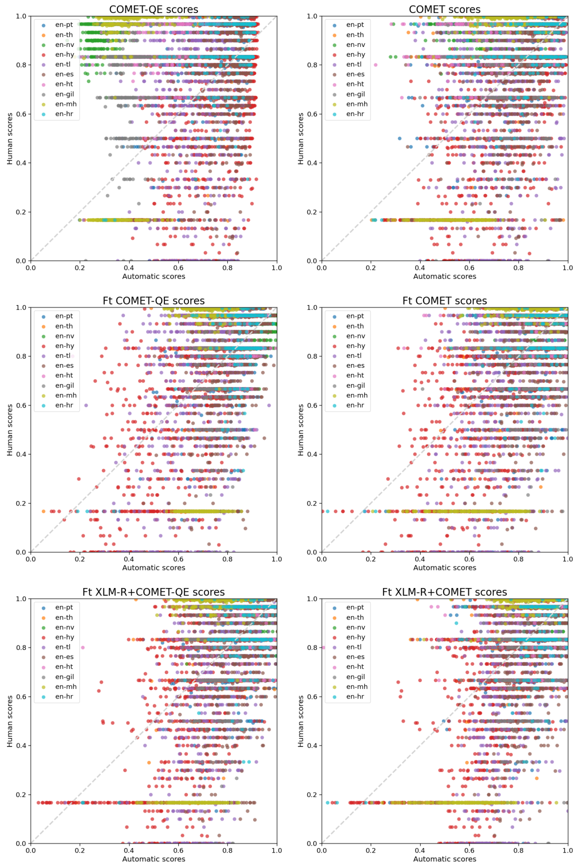

Figure 3: Human scores vs. automatic scores color-coded by language pair.